

hi

1:10 PM

Checkpoint Restored·

- Redo

GitHub Copilot

Hi! How can I help with your math flashcards project?

I am working on **Task 1** of my Math Flashcards VibeCoding lab.

Fix the string concatenation calculation bug in game/game_engine.py.

Currently compute_expected_answer() joins num_a and num_b as text.

For example, $13 + 5$ becomes 135 instead of 18.

Fix it so the answer is calculated correctly according to the active operator:

+ for addition

- for subtraction

* for multiplication

Do not change the existing UI or other game behavior.

Make the smallest clean change necessary.

Explain what you changed.

✅ Fixed

Updated game_engine.py:40-45 to calculate:

+ → addition

- → subtraction

* → multiplication

The previous string concatenation was replaced with direct arithmetic. No UI or other game behavior changed.

Verification

Validated all three operators:

$$13 + 5 = 18$$

$$13 - 5 = 8$$

$$13 * 5 = 65$$

Python diagnostics report **no errors**.

Task 2: Implement a per-question 10-second countdown timer in the Math Flashcards game.

Requirements:

- Start the timer whenever a new flashcard appears.
- Display a visible countdown timer bar below the active card.
- The bar should decrease as time passes.
- When 10 seconds expires before the user submits:
 - register the attempt as missed/incorrect,
 - reset the streak if applicable,
 - move to the next flashcard,
 - show a clear "TIME'S UP!" message.
- Reset the timer for every new question.

- Keep the existing UI and gameplay working.
- Do not break Task 1 arithmetic calculations.
- Make the smallest clean changes necessary.
- Explain which files you changed and why.

✅ Task 2 Implemented

Changed game_engine.py:

- Added a **10-second countdown** per question.
- Reset the timer whenever a new card appears.
- Added a visible decreasing timer bar below the card.
- Timeout increments the attempt, displays **“TIME’S UP!”**, and generates the next card.
- Preserved Task 1 arithmetic calculations.
- No streak system exists in the current project, so there was nothing to reset.

Verification

- Timeout, card reset, and attempt
- counting passed.
- Addition, subtraction, and multiplication checks passed.
- Pylance reports **no errors**.
- counting passed.
- Addition, subtraction, and multiplication checks passed.
- Pylance reports **no errors**.

Task 3: Add consecutive correct-answer streak multipliers to the Math Flashcards game.

Requirements:

- Track the number of consecutive correct answers.
- Increase the score reward when the player maintains a streak.

- Use escalating multipliers for consecutive correct answers.
- Reset the streak and multiplier to the baseline when:
 - the player gives a wrong answer, or
 - the timer expires.
- Clearly show the current streak and multiplier in the game UI.
- Keep the existing score and attempt tracking.
- Keep Task 1 arithmetic calculations working.
- Keep Task 2 countdown timer working.
- Make clean, minimal changes.
- Explain which files you changed and why.

✅ Task 3 Implemented

Updated game_engine.py:

Tracks consecutive correct answers.

Escalates rewards from 1× to 5×.

Displays streak and multiplier in the HUD.

Resets streak and multiplier after wrong answers or timeouts.

Preserves score, attempt tracking, timer, and arithmetic calculations.

Verification

Validated streak progression through 6 correct answers, both reset paths, the 10-second timer, and addition/subtraction/multiplication. Pylance reports **no errors**.

Task 4: Add division (/) to the Math Flashcards game.

Requirements:

- Add "/" to the available arithmetic operators.
- Generated division questions must always have a whole-number integer answer.
- There must be no remainder.
- Avoid division by zero.

- Keep +, -, and * working.
- Make sure the displayed problem and calculated answer are correct.
- Keep Task 1 arithmetic calculations working.
- Keep Task 2 countdown timer working.
- Keep Task 3 streak multipliers working.
- Make clean, minimal changes.
- Explain which files you changed and why.

✅ Task 4 Implemented

Updated game_engine.py:35-60:

- Added / to available operators.
- Generates division questions with a divisor from 2–7.
- Calculates the dividend as divisor × quotient, guaranteeing a whole-number answer.
- Prevents division by zero.
- Preserves existing +, -, and * behavior.
- Keeps Task 2 timer and Task 3 streak logic unchanged.

Verification

Validated **1,000 generated division questions** for:

- Nonzero divisors
- Integer answers

Correct displayed calculations

All arithmetic checks passed, and Pylance reports **no errors**.